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EXPLORING DATA PATTERNS WITH AUTOCORRELATION ANALYSIS

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AUTOCORRELATION ANALYSIS

- The autocorrelation analysis for different time lags of a variable is used **to identify time series data patterns** including components such as trend and seasonality.
- **Autocorrelation** : *The correlation between a variable lagged one or more periods and itself.*
- This is measured using the ***autocorrelation coefficient*** ρ_k and it's estimated by its ***sample autocorrelation coefficient*** r_k at lag *k* ; *k*=0,1,2,…

$$r_k = \frac{\sum_{t=k+1}^{n} (Y_t - \overline{Y})(Y_{t-k} - \overline{Y})}{\sum_{t=1}^{n} (Y_t - \overline{Y})^2} \quad k = 0, 1, 2, \ldots.$$

r_k = autocorrelation coefficient for a lag of *k* *periods*

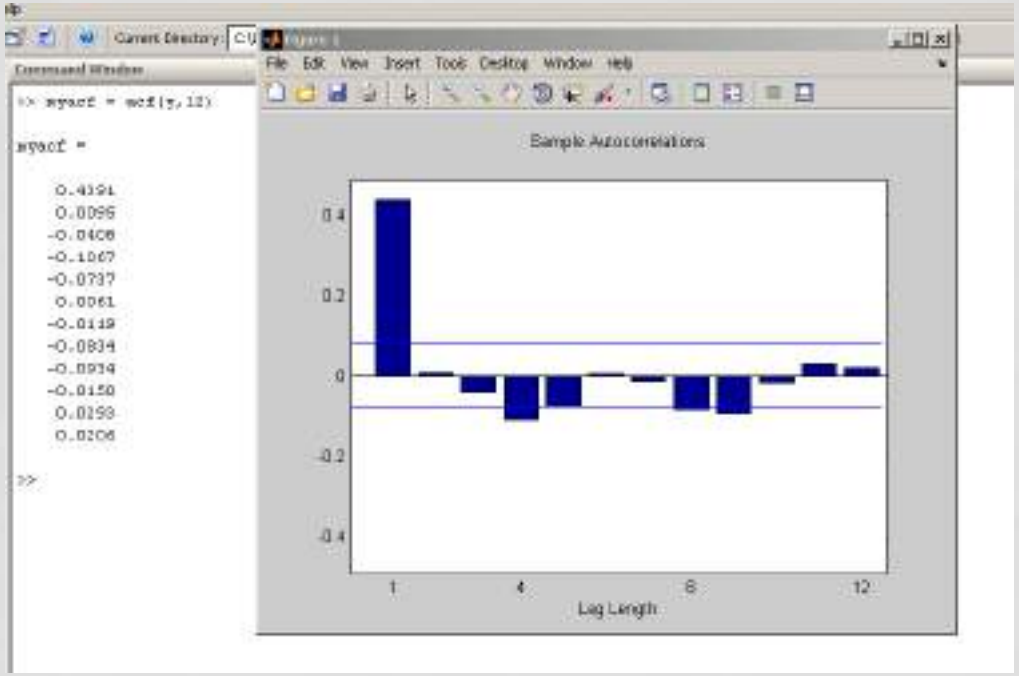
Y_t = observation in time period *t*

Y_{t-k} = observation at time period *t-k*

$\overline{Y}$ = mean of the values of the series

CORRELOGRAM

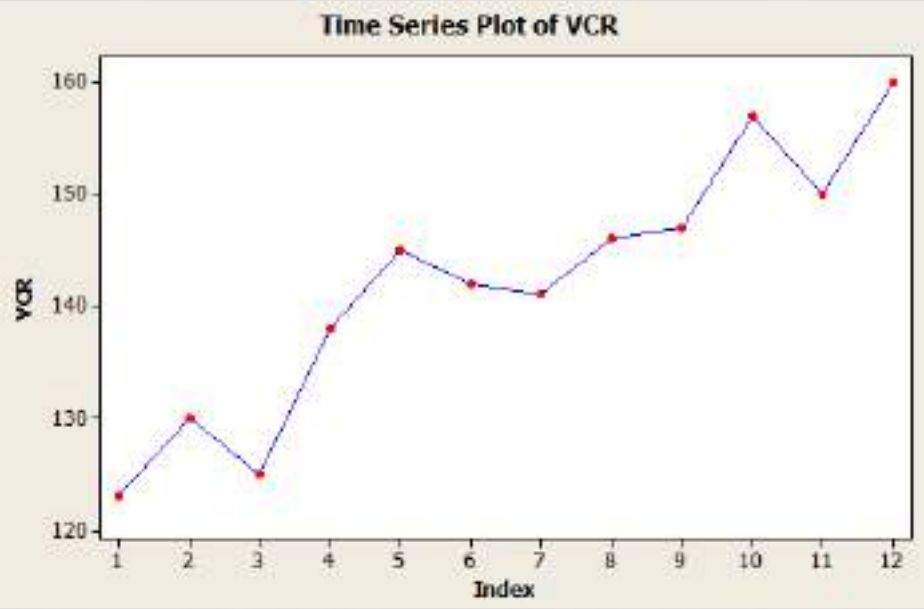
- Autocorrelation analysis can be done by demonstrating the ***correlogram*** or *autocorrelation function*.
- The *correlogram* or *autocorrelation function* is a **graph of the autocorrelations for various lags of a time series**.



EXAMPLE 1

Harry Vernon has collected data on the number of VCRs sold last year for Vernon's Music Store.

We need to know lag 1 and lag 2 autocorrelation coefficients (r_1 and r_2)



Month	# of VCRs
January	123
February	130
March	125
April	138
May	145
June	142
July	141
August	146
September	147
October	157
November	150
December	160

SOLUTION

Time t	Month	Original Data Y_t	Y Lagged One Period Y_{t-1}	Y Lagged Two Periods Y_{t-2}
1	January	123		
2	February	130	123	
3	March	125	130	123
4	April	138	125	130
5	May	145	138	125
6	June	142	145	138
7	July	141	142	145
8	August	146	141	142
9	September	147	146	141
10	October	157	147	146
11	November	150	157	147
12	December	160	150	157

$$\bar{Y} = \frac{1,704}{12} = 142$$

LAG 1

$$r_1 = \frac{\sum_{t=1+1}^{n}(Y_t - \bar{Y})(Y_{t-1} - \bar{Y})}{\sum_{t=1}^{n}(Y_t - \bar{Y})^2} = \frac{843}{1474} = 0.572$$

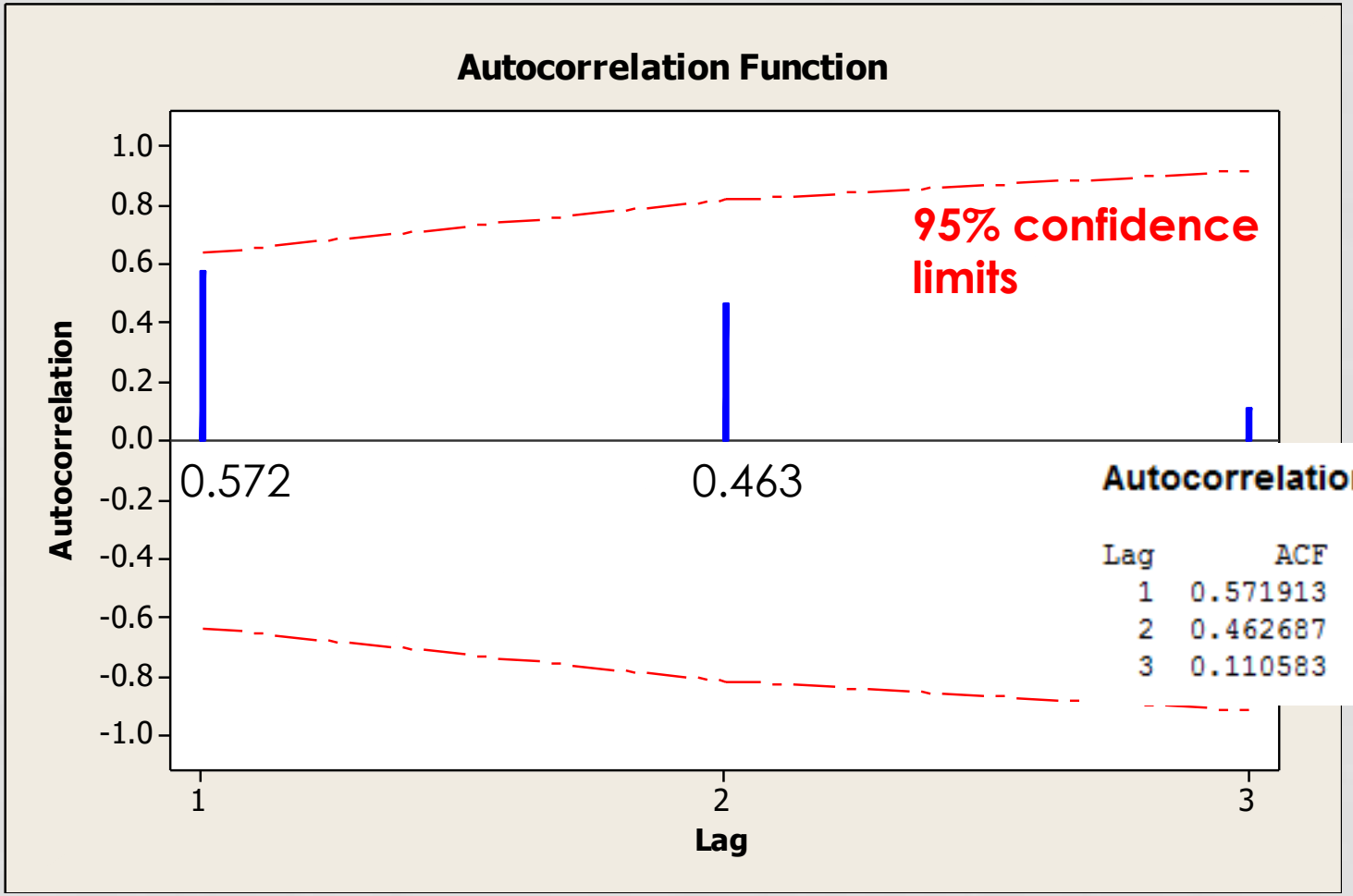
Time, t	Y_t	Y_{t-1}	$(Y_t - \bar{Y})$	$(Y_{t-1} - \bar{Y})$	$(Y_t - \bar{Y})^2$	$(Y_t - \bar{Y})(Y_{t-1} - \bar{Y})$
1	123	—	−19	—	361	—
2	130	123	−12	−19	144	228
3	125	130	−17	−12	289	204
4	138	125	−4	−17	16	68
5	145	138	3	−4	9	−12
6	142	145	0	3	0	0
7	141	142	−1	0	1	0
8	146	141	4	−1	16	−4
9	147	146	5	4	25	20
10	157	147	15	5	225	75
11	150	157	8	15	64	120
12	160	150	18	8	324	144
Total	1,704		0		1,474	843

LAG 2

$$r_2 = \frac{\sum_{t=2+1}^{n} (Y_t - \bar{Y})(Y_{t-2} - \bar{Y})}{\sum_{t=1}^{n} (Y_t - \bar{Y})^2} = \frac{682}{1474} = 0.463$$

Time, t	Y_t	Y_{t-2}	$(Y_t-\bar{Y})^2$	$(Y_{t-2}-\bar{Y})$	$(Y_t-\bar{Y})$	$(Y_t-\bar{Y})(Y_{t-2}-\bar{Y})$
1	123	-	361	-	-19	-
2	130	-	144	-	-12	-
3	125	123	289	-19	-17	323
4	138	130	16	-12	-4	48
5	145	125	9	-17	3	-51
6	142	138	0	-4	0	0
7	141	145	1	3	-1	-3
8	146	142	16	0	4	0
9	147	141	25	-1	5	-5
10	157	146	225	4	15	60
11	150	147	64	5	8	40
12	160	157	324	15	18	270
Total	1,704		1,474			682

CORRELOGRAM BY MINITAB



Autocorrelation Function: VCR

Lag	ACF	T	LBQ
1	0.571913	1.98	5.00
2	0.462687	1.25	8.59
3	0.110583	0.27	8.82

AUTOCORRELATION COEFFICIENT TO EXPLORE DATA PATTERN

- Autocorrelation coefficients for different time lags for a variable can be used to answer the following questions about a time series:

  - **1. Are the data random?**
  - **2. Do the data have a trend?**
  - **3. Are the data stationary?**
  - **4. Are the data seasonal?**

AUTOCORRELATION COEFFICIENT TO EXPLORE DATA PATTERN (2)

- If a series is **random**, the autocorrelations between *Y*, and *Yt-k* for any lag k are **close to zero**. The successive values of a time series are **not related** to each other.

- If a series has a **trend**, successive observations are **highly correlated** and the autocorrelation coefficients are **typically significantly different from zero for the first several time lags and then gradually drop toward zero as the number of lags increases.**
- The autocorrelation coefficient for time lag 1: is large (close to 1), for time lag 2: is also large but smaller than for time lag 1.

- If a series has a **seasonal** pattern, a significant autocorrelation coefficient will **occur at the seasonal time lag or multiples of the seasonal lag**.
- The seasonal lag is 4 for quarterly data and 12 for monthly data.

IS AN AUTOCORRELATION COEFFICIENT SIGNIFICANTLY DIFFERENT FROM ZERO?

- The autocorrelation coefficients of random data have an approximate normal sampling distribution (Quenouille, 1949).
- At a specified confidence level, a series can be considered **random if the autocorrelation coefficients are within the interval [0 ± *t* x *SE*(r_k)]**, (z instead of *t* for large samples).
- The following *t* statistic can be used:

$$t = \frac{r_k}{SE(r_k)}$$

t = appropriate percentage point of a t distribution

$$SE(r_k) = \sqrt{\frac{1 + 2\sum_{i=1}^{k-1} r_i^2}{n}}$$

SE(r_k) = standard error of the autocorrelation at time lag k

SE(r_1) = 1 / √n

r_i = the autocorrelation at time lag *i*.

k = the time lag

n = the number of observations in the time series

HYPOTHESES TEST

- At significant level α = 0.05: the critical values ± 2.2 are the *t* upper and lower points for *n-1* = 11 degrees of freedom.

$$|t| \geq t_{\frac{\alpha}{2},n-1}$$

- The null and alternative hypotheses :

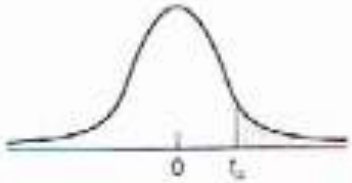
$$H_0 : \rho_k = 0$$
$$H_1 : \rho_k \neq 0$$

- Decision Rule:

If *t* < -2.2 or *t* > 2.2, reject H_0 , conclude that the lag 1 autocorrelation is significantly different from 0

- Note: t is given directly in the Minitab output under the heading T.

T DISTRIBUTION TABLE



df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	3.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.529	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
inf.	1.282	1.645	1.960	2.326	2.576

HYPOTHESES TEST (2)

$SE(r_1) = \sqrt{1/12} = .289$

$$SE(r_2) = \sqrt{\frac{1+2\sum_{i=1}^{2-1} r_i^2}{n}} = \sqrt{\frac{1+2(.572)^2}{12}} = .371$$

$$t = \frac{r_1 - \rho_1}{SE(r_1)} = \frac{.572}{.289} = 1.98$$

Ho: $\rho_1 = 0$ cannot be rejected because $-2.2 < 1.98 < 2.2$.

$$t = \frac{r_2}{SE(r_2)} = \frac{.463}{.371} = 1.25$$

Ho: $\rho_2 = 0$ cannot be rejected because $-2.2 < 1.25 < 2.2$.

then the time series data is random

95% CONFIDENCE LIMITS

- An alternative way to check for significant autocorrelation is to construct, say, **95% confidence limits centered at 0**.
- These limits for lags 1 and 2 are given by

$$\text{lag 1: } 0 \pm t_{.025} \times SE(r_1) \quad \text{or} \quad 0 \pm 2.2(.289) \rightarrow (-.636, .636)$$

$$\text{lag 2: } 0 \pm t_{.025} \times SE(r_2) \quad \text{or} \quad 0 \pm 2.2(.371) \rightarrow (-.816, .816)$$

- Autocorrelation significantly different from 0 is indicated whenever a value for ***r* falls outside the corresponding confidence limits.**
- The 95% confidence limits are shown in correlogram by the **dashed lines** in the graphical display of the autocorrelation function.

LBQ

- **The Modified Box-Pierce Q statistic (developed by: Ljung, and Box) "LBQ"**
- A **portmanteau test**: to see whether a set (ex: the first 10 r_k values) is significantly different from a set in which all 10 values are zero.

$$Q = n(n+2)\sum_{k=1}^{m}\frac{r_k^2}{n-k}$$

n= number of observations

k= the time lag

m= number of time lags to be considered

r_k = k^{th} autocorrelation coefficient lagged *k* time periods

LBQ (2)

- The value of Q can be compared with the chi-square with *m* degrees of freedom.

$$Q \geq x^2_{\alpha,m}$$

Evidence against the null hypothesis

- For example, at 5 % level of significant, the time series **data are random** (the null hypothesis cannot be rejected) if Q for a subset of 10 autocorrelations is **less than** $x^2_{0.05,10} = 18.31$

χ^2 DISTRIBUTION TABLE

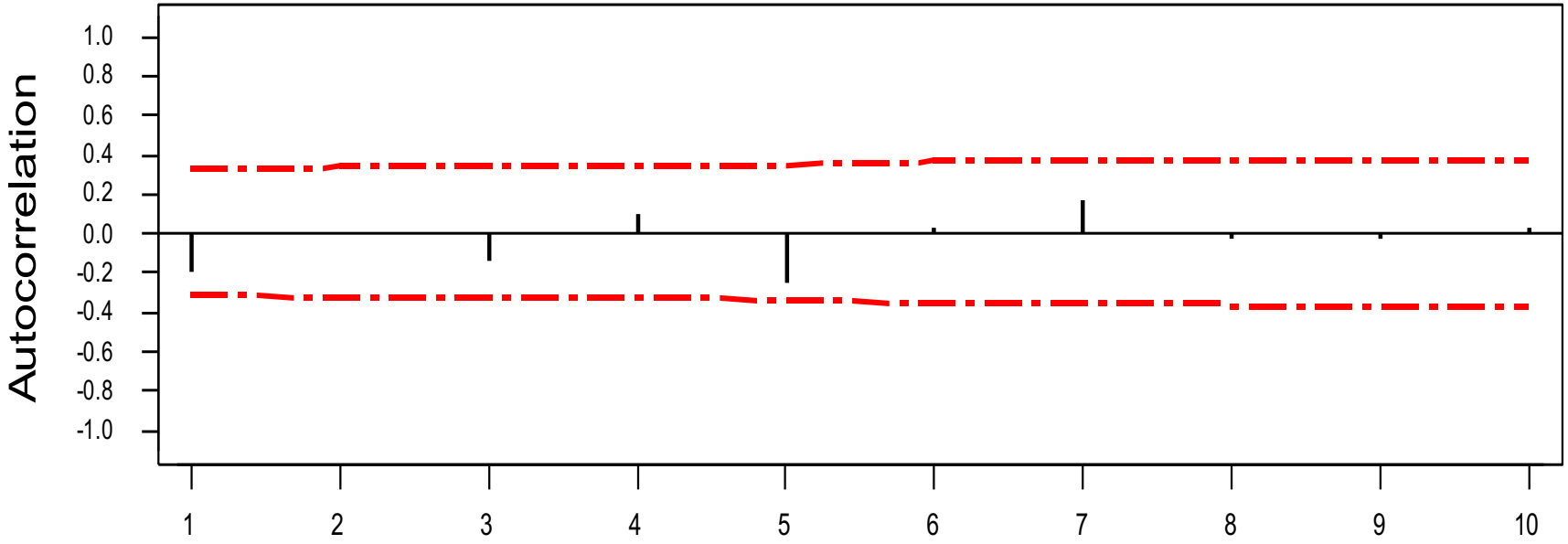
df	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	2.70554	3.84146	5.02389	6.63490	7.87944
2	4.60517	5.99147	7.37776	9.21034	10.5966
3	6.25139	7.81473	9.34840	11.3449	12.8381
4	7.77944	9.48773	11.1433	13.2767	14.8602
5	9.23635	11.0705	12.8325	15.0863	16.7496
6	10.6446	12.5916	14.4494	16.8119	18.5476
7	12.0170	14.0671	16.0128	18.4753	20.2777
8	13.3616	15.5073	17.5346	20.0902	21.9550
9	14.6837	16.9190	19.0228	21.6660	23.5893
10	15.9871	18.3070	20.4831	23.2093	25.1882
11	17.2750	19.6751	21.9200	24.7250	26.7569
12	18.5494	21.0261	23.3367	26.2170	28.2995
13	19.8119	22.3621	24.7356	27.6883	29.8194
14	21.0642	23.6848	26.1190	29.1413	31.3193
15	22.3072	24.9958	27.4884	30.5779	32.8013
16	23.5418	26.2962	28.8454	31.9999	34.2672
17	24.7690	27.5871	30.1910	33.4087	35.7185
18	25.9894	28.8693	31.5264	34.8053	37.1564
19	27.2036	30.1435	32.8523	36.1908	38.5822
20	28.4120	31.4104	34.1696	37.5662	39.9968
21	29.6151	32.6705	35.4789	38.9321	41.4010
22	30.8133	33.9244	36.7807	40.2894	42.7956
23	32.0069	35.1725	38.0757	41.6384	44.1813
24	33.1963	36.4151	39.3641	42.9798	45.5585
25	34.3816	37.6525	40.6465	44.3141	46.9278
26	35.5631	38.8852	41.9232	45.6417	48.2899
27	36.7412	40.1133	43.1944	46.9630	49.6449
28	37.9159	41.3372	44.4607	48.2782	50.9933
29	39.0875	42.5569	45.7222	49.5879	52.3356
30	40.2560	43.7729	46.9792	50.8922	53.6720
40	51.8050	55.7585	59.3417	63.6907	66.7659
50	63.1671	67.5048	71.4202	76.1539	79.4900
60	74.3970	79.0819	83.2976	88.3794	91.9517
70	85.5271	90.5312	95.0231	100.425	104.215
80	96.5782	101.879	106.629	112.329	116.321
90	107.565	113.145	118.136	124.116	128.299
100	118.498	124.342	129.561	135.807	140.169

EXAMPLE 2

Time series of 40 random numbers

t	Y_t	t	Y_t	t	Y_t	t	Y_t
1	343	11	946	21	704	31	555
2	574	12	142	22	291	32	476
3	879	13	477	23	43	33	612
4	728	14	452	24	118	34	574
5	37	15	727	25	682	35	518
6	227	16	147	26	577	36	296
7	613	17	199	27	834	37	970
8	157	18	744	28	981	38	204
9	571	19	627	29	263	39	616
10	72	20	122	30	424	40	97

Autocorrelation Function



Lag	Corr	T	LBQ	Lag	Corr	T	LBQ
1	-0.19	-1.21	1.57	8	-0.03	-0.15	7.67
2	-0.01	-0.04	1.58	9	-0.03	-0.18	7.73
3	-0.15	-0.89	2.53	10	0.02	0.12	7.75
4	0.10	0.63	3.04				
5	-0.25	-1.50	6.13				
6	0.03	0.16	6.17				
7	0.17	0.95	7.63				

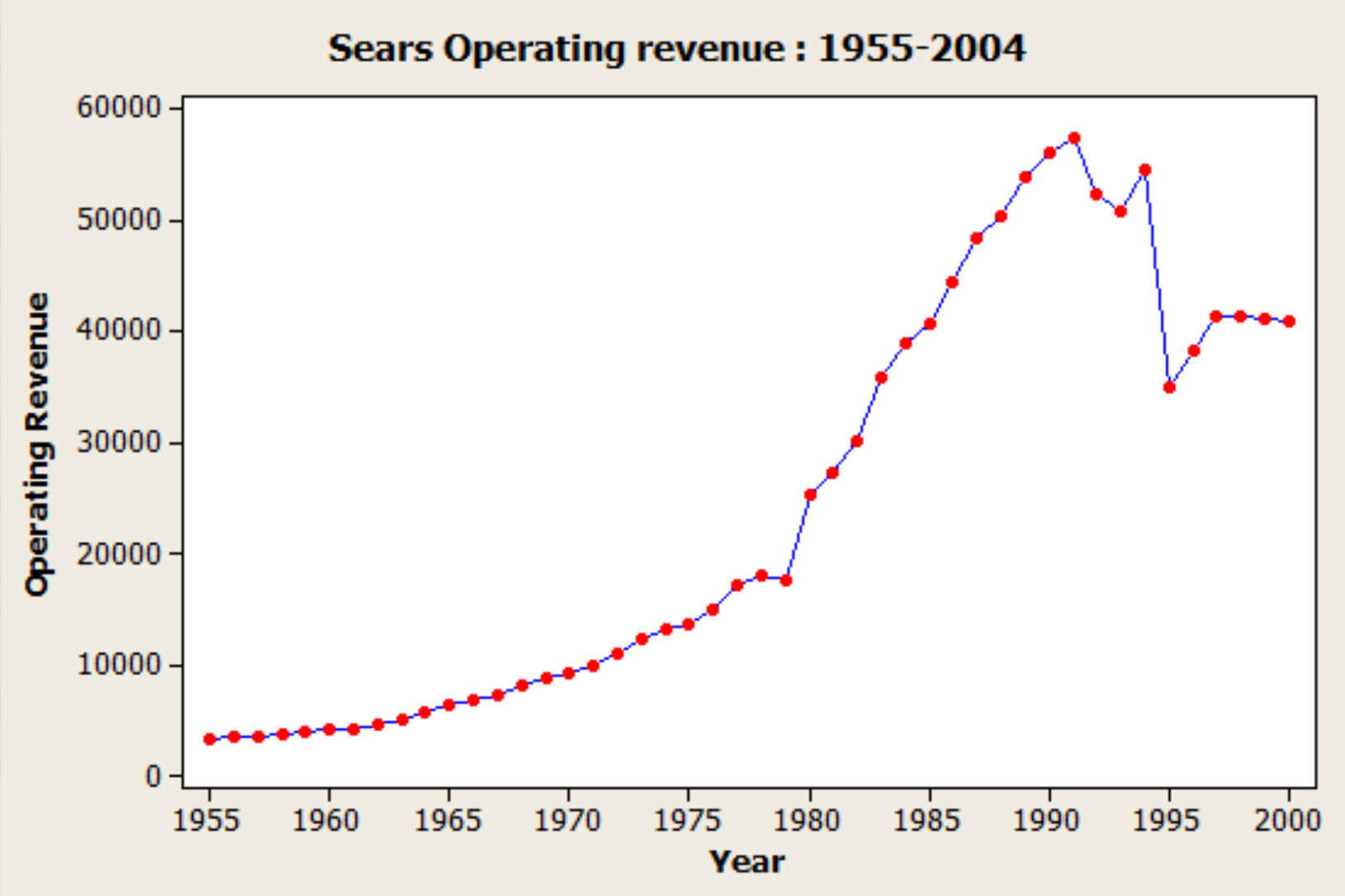
Q statistic for *m*= 10 time lags

7.75 < 18.31 Conclusion: the series is random

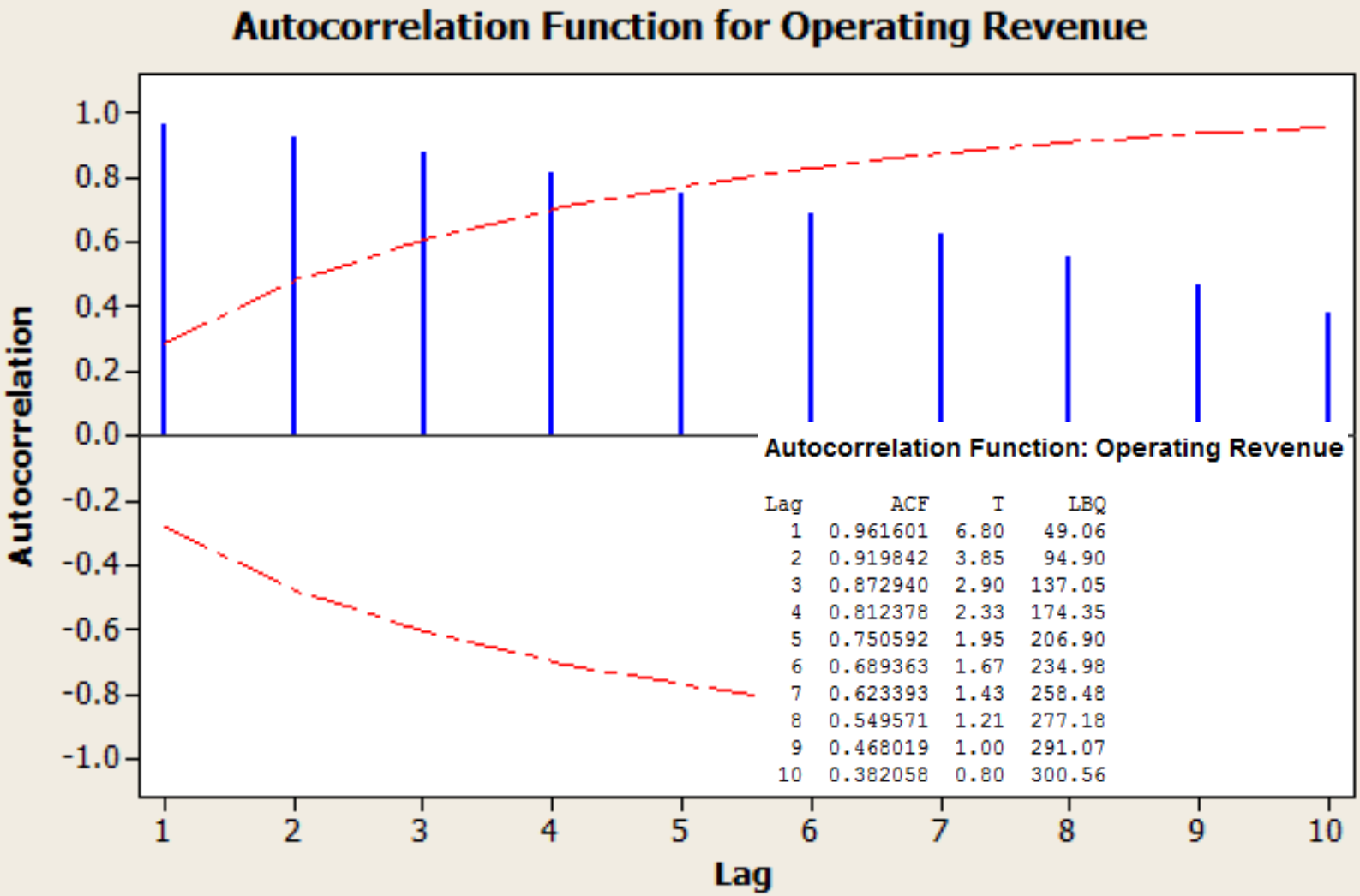
An analyst for sears company is assigned the task of forecasting operating revenue for 2005. she gathers the data for the years 1955 to 2004 as shown in table below

EXAMPLE 3

Year	Y_t	Year	Y_t	Year	Y_t	Year	Y_t	Year	Y_t
1955	**3307**	1966	**6769**	1977	**17224**	1988	**50251**	1999	**41071**
1956	**3556**	1967	**7296**	1978	**17946**	1989	**53794**	2000	**40937**
1957	**3601**	1968	**8178**	1979	**17514**	1990	**55972**	2001	**36151**
1958	**3721**	1969	**8844**	1980	**25195**	1991	**57242**	2002	**30762**
1959	**4036**	1970	**9251**	1981	**27357**	1992	**52345**	2003	**23253**
1960	**4134**	1971	**10006**	1982	**30020**	1993	**50838**	2004	**19701**
1961	**4268**	1972	**10991**	1983	**35883**	1994	**54559**		
1962	**4578**	1973	**12306**	1984	**38828**	1995	**34925**		
1963	**5093**	1974	**13101**	1985	**40715**	1996	**38236**		
1964	**5716**	1975	**13639**	1986	**44282**	1997	**41296**		
1965	**6357**	1976	**14950**	1987	**48440**	1998	41322		



TIME SERIES PLOT



Autocorrelation Function: Operating Revenue

Lag	ACF	T	LBQ
1	0.961601	6.80	49.06
2	0.919842	3.85	94.90
3	0.872940	2.90	137.05
4	0.812378	2.33	174.35
5	0.750592	1.95	206.90
6	0.689363	1.67	234.98
7	0.623393	1.43	258.48
8	0.549571	1.21	277.18
9	0.468019	1.00	291.07
10	0.382058	0.80	300.56

CORRELOGRAM

ANALYSIS

A 95% confidence interval for the autocorrelation coefficients at time lag 1 using $0 \pm Z_{025}(\sqrt{1/50})$.

$Z_{.025}$ -> see $t_{.025,}$
df = n – 1 = 50 – 1 = 49
= 1.96

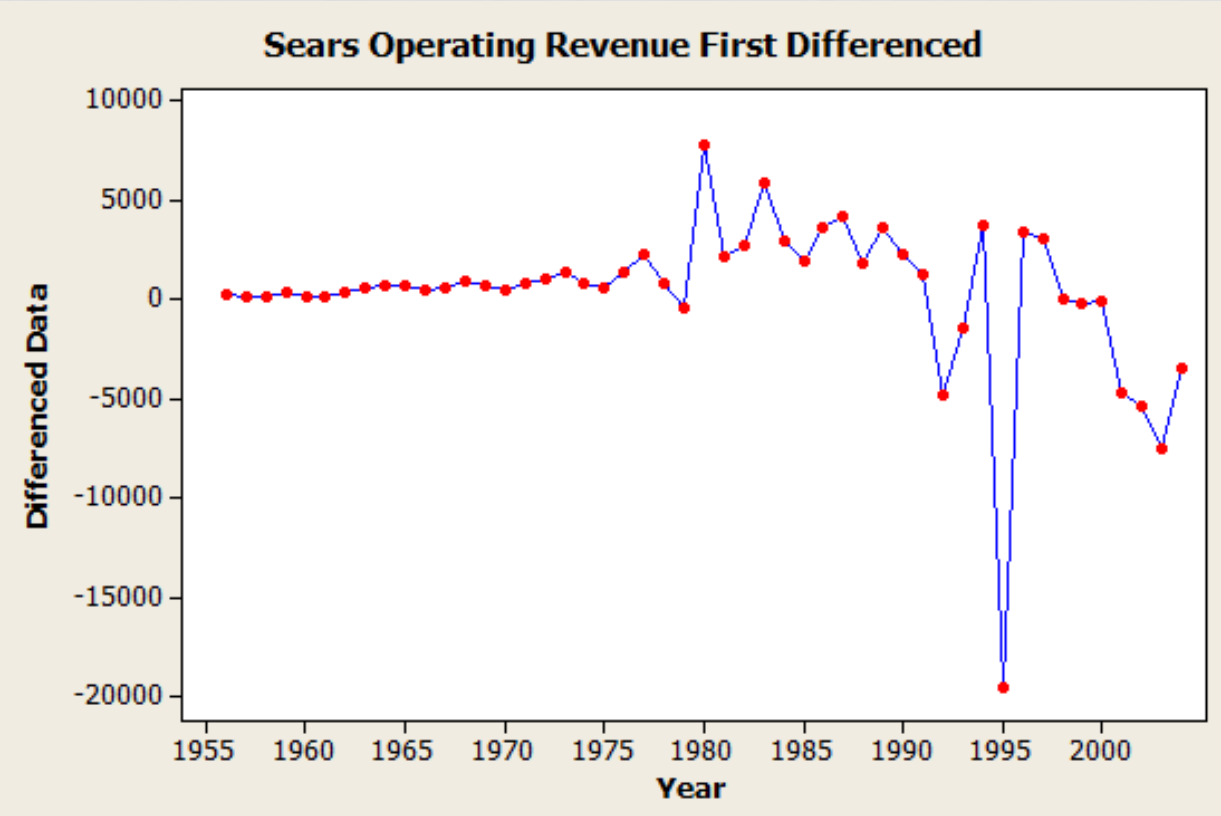
0 ± 0.277

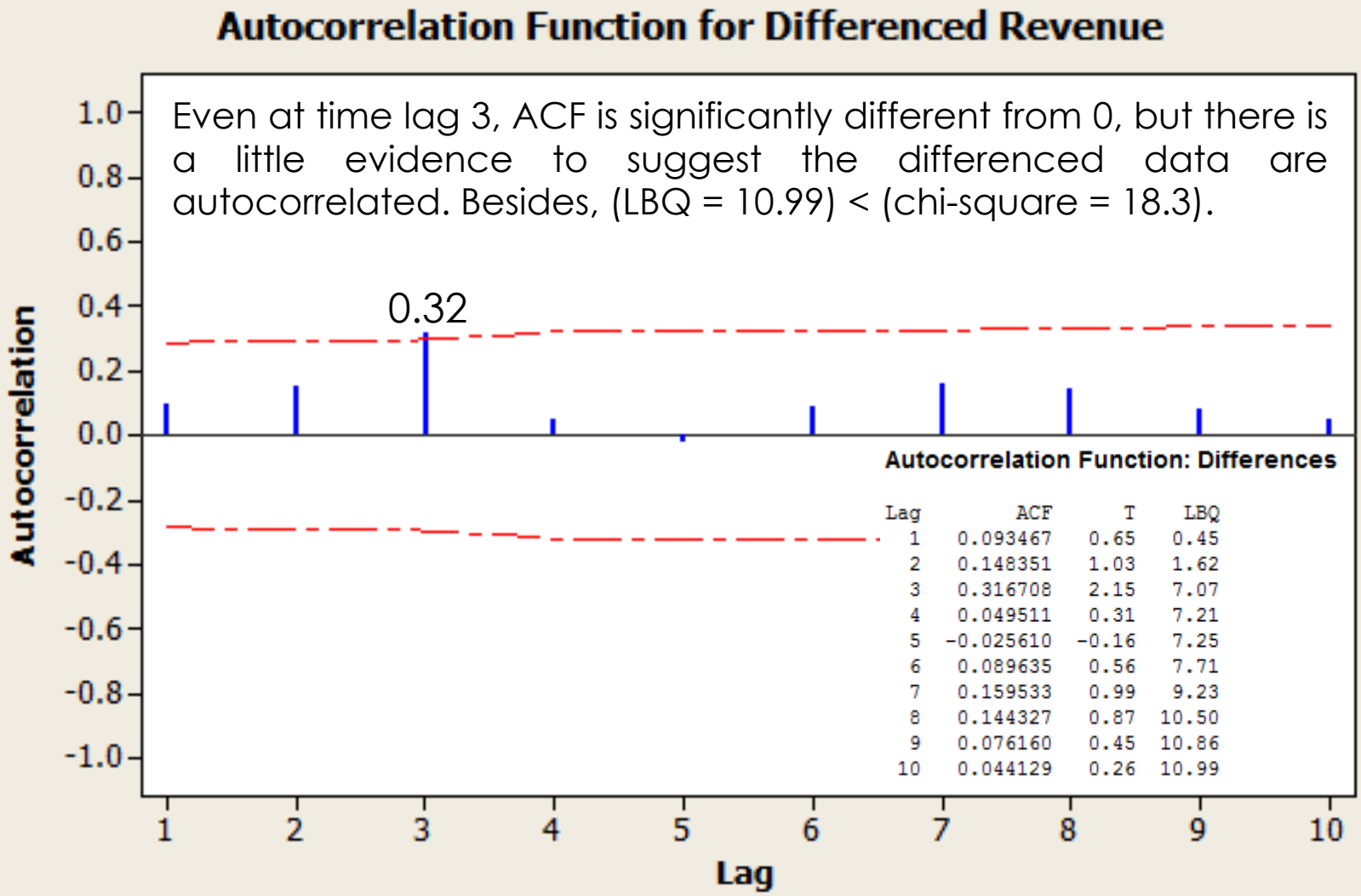
Time lags are significantly different from zero (.96, .92. and .87) and that the values then gradually drop to zero

(LBQ = 300.56) > (chi-square = 18.3)
Conclusion : Reject null hyphothesis, autocorrelations for the first 10 lags as a group are significantly different from zero, data are highly correlated, exhibit **trend**like behavior

DATA DIFFERENCING

- **A time series can be differenced to remove the trend and to create a stationary series.**

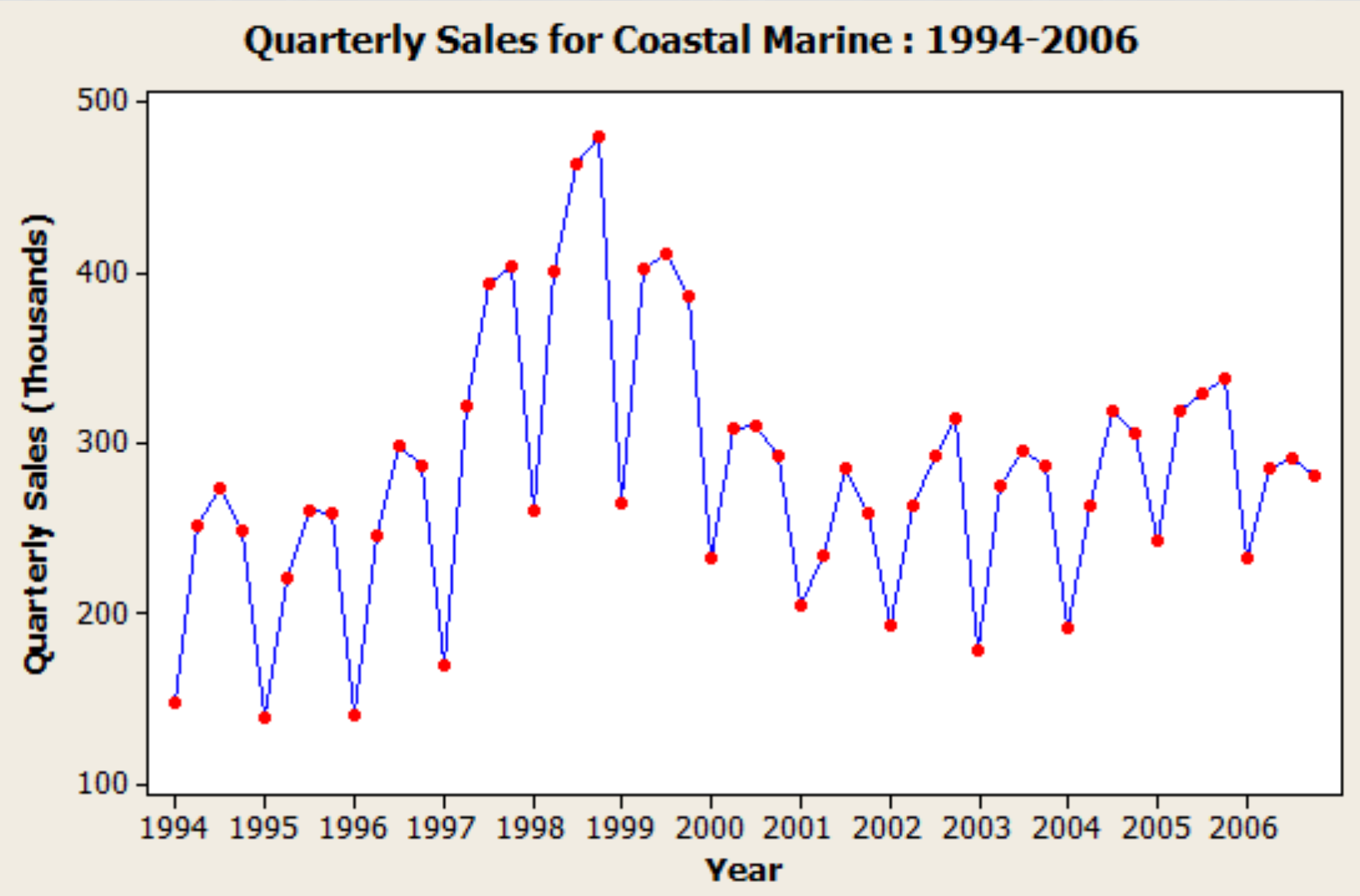




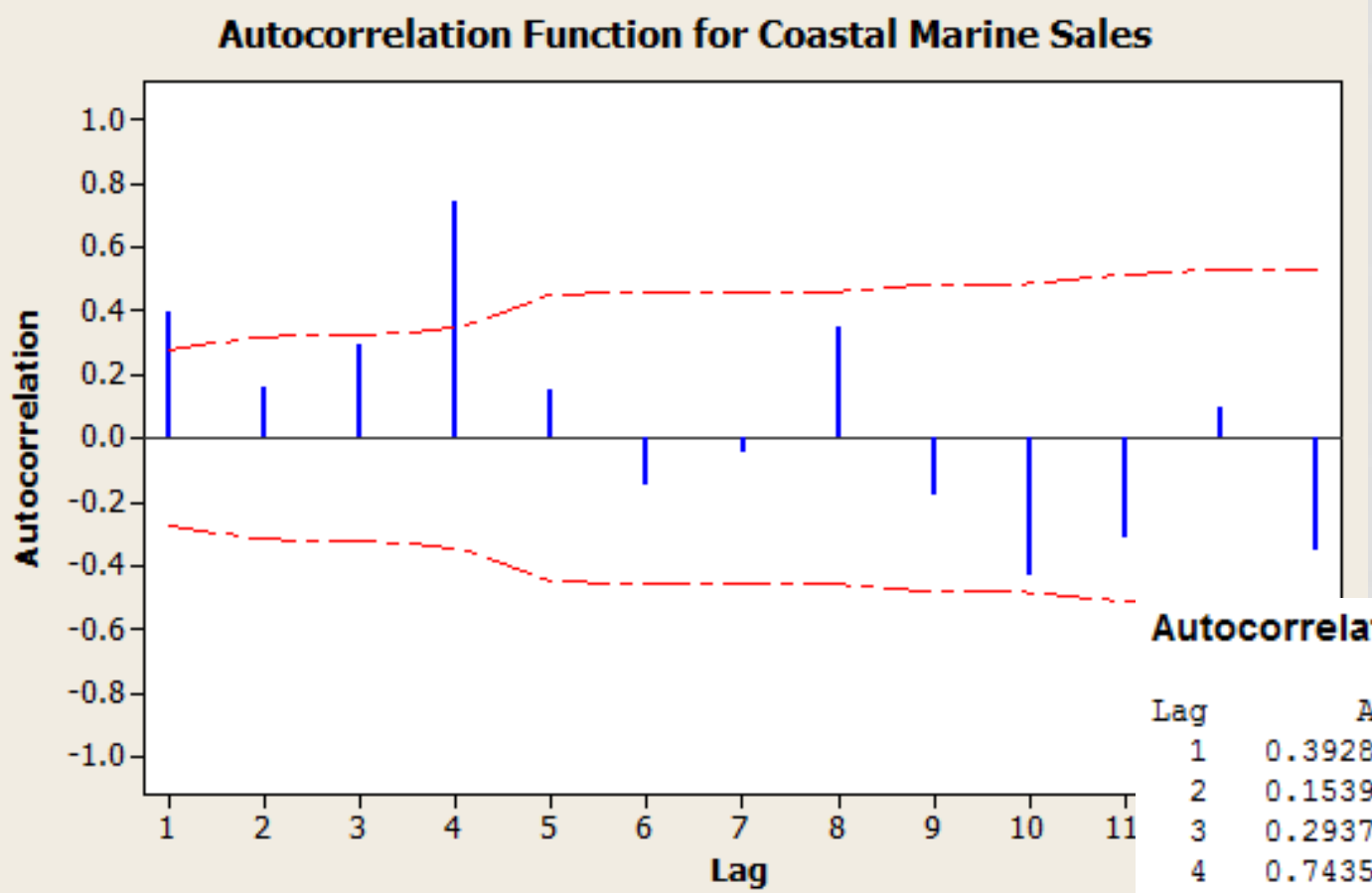
CORRELOGRAM

EXAMPLE 4 : An analyst for outboard marine corporation always felt that sales were seasonal. He gathers the data shown in the table below for the quarterly sales of outboard marine corporation from 1994 to 2006

Year	December 31	March 31	June 30	September 30
1994	147.6	251.8	273.1	249.1
1995	139.3	221.2	260.2	259.5
1996	140.5	245.5	298.8	287.0
1997	168.8	322.6	393.5	404.3
1998	259.7	401.1	464.6	497.7
1999	264.4	402.6	411.3	385.9
2000	232.7	309.2	310.7	293.0
2001	205.1	234.4	285.4	258.7
2002	193.2	263.7	292.5	315.2
2003	178.3	274.5	295.4	286.4
2004	190.8	263.5	318.8	305.5
2005	242.6	318.8	329.6	338.2
2006	232.1	285.6	291.0	281.4



TIME SERIES PLOT



$r_1 = 0.39 > 0.272$
$r_4 = 0.74 > 0.333$
Autocorrelation coefficients at time lags 1 and 4 are significantly different from zero, Sales are **seasonal** on quarterly basis.

Autocorrelation Function: Sales

Lag	ACF	T	LBQ
1	0.392843	2.83	8.50
2	0.153948	0.97	9.83
3	0.293739	1.82	14.77
4	0.743533	4.34	47.11
5	0.150619	0.67	48.47
6	-0.153014	-0.67	49.90
7	-0.047049	-0.21	50.04
8	0.346979	1.51	57.72
9	-0.182597	-0.76	59.90
10	-0.434737	-1.80	72.53
11	-0.315090	-1.23	79.33
12	0.091175	0.35	79.92
13	-0.353272	-1.34	88.90

A 95% confidence interval for the autocorrelation coefficients at time lag 1 using $0 \pm Z_{025}(\sqrt{1/52})$.

$Z_{.025}$ -> see $t_{.025}$, df = n – 1 = 50 – 1 = 49 = 1.96

0 ± 0.272

Method	Pattern of Data	Time Horizon	Type of Model	Minimal Data Requirements	
				Nonseasonal	Seasonal
Naïve	ST , T , S	S	TS	1	
Simple averages	ST	S	TS	30	
Moving averages	ST	S	TS	4-20	
Single Exponential smoothing	ST	S	TS	2	
Linear (Double) exponential smoothing (Holt's)	T	S	TS	3	
Quadratic exponential smoothing	T	S	TS	4	
Seasonal exponential smoothing (Winter's)	S	S	TS		2 x s
Adaptive filtering	S	S	TS		5 x s
Simple regression	T	I	C	10	
Multiple regression	C , S	I	C	10 x V	
Classical decomposition	S	S	TS		5 x s
Exponential trend models	T	I , L	TS	10	
S-curve fitting	T	I , L	TS	10	
Gompertz models	T	I , L	TS	10	
Growth curves	T	I , L	TS	10	
Census X-12	S	S	TS		6 x s
ARIMA (Box-Jenkins)	ST , T , C , S	S	TS	24	3 x s
Lading indicators	C	S	C	24	
Econometric models	C	S	C	30	
Time series multiple regression	T , S	I , L	C		6 x s

Pattern of data: ST, stationary; T, trend; S, seasonal; C, cyclical.
Time horizon: S, short term (less than three months); I, intermediate; L, long term
Type of model: TS, time series; C, causal.
Seasonal: s, length of seasonality.
Variable: V, number of variables.